Li, Yicheng

Ph.D. Candidate in the Department of Statistics and Data Science, Tsinghua University

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Education and Research Experience

2022 – Present **Ph.D. in Statistics**.
Department of Statistics and Data Science, Tsinghua University, Beijing, China.
Advisor: Professor Qian Lin.

Fall 2025 **Visiting Student**, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, Philadelphia, USA.
Host: Professor Tony Cai.

2018 – 2022 **B.Sc. in Pure and Applied Mathematics**.
Department of Mathematical Sciences, Tsinghua University, Beijing, China.
Minor in Data Science and Technology.
Outstanding Graduate (top 3/111).

Research Interest

deep learning theory, differential privacy, feature learning, kernel methods

Research Publications

Journal Articles

1. *Optimal Rates of Kernel Ridge Regression under Source Condition in Large Dimensions*
Haobo Zhang, **Yicheng Li**, Weihao Lu, and Qian Lin
Journal of Machine Learning Research, vol. 26, no. 219, pp. 1–63, 2025
URL: http://jmlr.org/papers/v26/23-1679.html

2. *On the Eigenvalue Decay Rates of a Class of Neural-Network Related Kernel Functions Defined on General Domains*
Yicheng Li, Zixiong Yu, Guhan Chen, and Qian Lin
Journal of Machine Learning Research, vol. 25, no. 82, pp. 1–47, 2024
URL: http://jmlr.org/papers/v25/23-0866.html

3. *Kernel Interpolation Generalizes Poorly*
Yicheng Li, Haobo Zhang, and Qian Lin
Biometrika, vol. 111, no. 2, pp. 715–722, 2024
URL: https://academic.oup.com/biomet/article-abstract/111/2/715/7238206

4. *On the Optimality of Misspecified Spectral Algorithms*
Haobo Zhang, **Yicheng Li** (Co-first author), and Qian Lin
Journal of Machine Learning Research, vol. 25, no. 188, pp. 1–50, 2024
URL: https://www.jmlr.org/papers/v25/23-0383.html

Conference Proceedings

1. *Functional Virtual Adversarial Training for Semi-Supervised Time Series Classification*
Qingyi Pan and **Yicheng Li** (Corresponding author)
(accepted by NeurIPS 2025), Sep. 2025.
URL: https://openreview.net/forum?id=hYjlAA6wXT

2. *On the Impacts of the Random Initialization in the Neural Tangent Kernel Theory*

Guhan Chen, **Yicheng Li**, and Qian Lin
The Thirty-eighth Annual Conference on Neural Information Processing Systems, Sep. 2024.
URL: https://openreview.net/forum?id=ni3Ud2BV3G

3. *Improving Adaptivity via Over-Parameterization in Sequence Models*
Yicheng Li and Qian Lin
The Thirty-eighth Annual Conference on Neural Information Processing Systems, Sep. 2024.
URL: https://openreview.net/forum?id=UfLH4T676K

4. *On the Saturation Effects of Spectral Algorithms in Large Dimensions*
Weihao Lu, Haobo Zhang, **Yicheng Li**, and Qian Lin
The Thirty-eighth Annual Conference on Neural Information Processing Systems, Sep. 2024.
URL: https://openreview.net/forum?id=kJzecLYsRi

5. *On the Asymptotic Learning Curves of Kernel Ridge Regression under Power-Law Decay*
Yicheng Li, Haobo Zhang, and Qian Lin
Thirty-Seventh Conference on Neural Information Processing Systems, 2023.
URL: https://openreview.net/forum?id=E4P5kVSKlT

6. *On the Saturation Effect of Kernel Ridge Regression*
Yicheng Li, Haobo Zhang, and Qian Lin
International Conference on Learning Representations, Feb. 2023.
URL: https://openreview.net/forum?id=tFvr-kYWs_Y

7. *On the Optimality of Misspecified Kernel Ridge Regression*
Haobo Zhang, **Yicheng Li**, Weihao Lu, and Qian Lin
International Conference on Machine Learning, 2023.
URL: https://openreview.net/forum?id=Kg2al3GXBR

Preprints and Ongoing Works

1. *Differentially Private Functional Linear regression: Optimal Algorithms and Theoretical Guarantees*
Yicheng Li and T. Tony Cai
Sep. 2025. Submitted.

2. *Minimax and Adaptive Covariance Matrix Estimation under Differential Privacy*
Yicheng Li and T. Tony Cai
Nov. 2025. Ongoing Work.

3. *Diagonal Over-Parameterization in Reproducing Kernel Hilbert Spaces as an Adaptive Feature Model: Generalization and Adaptivity*
Yicheng Li and Qian Lin
Jan. 2025. arXiv: 2501.08679 (cs) URL: http://arxiv.org/abs/2501.08679.

4. *Several Supporting Evidences for the Adaptive Feature Program*
Yicheng Li and Qian Lin
Nov. 2025. arXiv: 2511.09425 (cs) DOI: 10.48550/arXiv.2511.09425.

5. *Generalization Error Curves for Analytic Spectral Algorithms under Power-Law Decay*
Yicheng Li, Weiye Gan, Zuoqiang Shi, and Qian Lin
Jan. 2024. arXiv: 2401.01599 (cs, math, stat) URL: http://arxiv.org/abs/2401.01599.

6. *Towards a Statistical Understanding of Neural Networks: Beyond the Neural Tangent Kernel Theories*
Haobo Zhang, Jianfa Lai, **Yicheng Li**, Qian Lin, and Jun S. Liu
Dec. 2024. arXiv: 2412.18756 (cs) URL: http://arxiv.org/abs/2412.18756.

7. *Optimal Rate of Kernel Regression in Large Dimensions*
Weihao Lu, Haobo Zhang, **Yicheng Li**, Manyun Xu, and Qian Lin
Sep. 2023. arXiv: 2309.04268 (cs, math, stat) URL: http://arxiv.org/abs/2309.04268.

Seminar & Conference Presentations

2025
- Invited Talk, *Diagonal Over-parameterization in Reproducing Kernel Hilbert Spaces as an Adaptive Feature Model: Generalization and Adaptivity*, 2025 ICSA China Conference.

2024
- Contributed Talk, *Generalization error curves for analytic spectral algorithms under power-law decay*, 8th Academic Symposium of the High-Dimensional Data Statistics Section of the Chinese Society of Statistical Research, Nanjing, April 2024.
- Contributed Poster, *Generalization error curves for analytic spectral algorithms under power-law decay*, Conference "Statistics in the Age of AI", May 2024, George Washington University, Washington DC, USA.

2023
- Speaker, Workshop on Machine Learning, August 2023, Hangzhou, China.
- Poster Presentation,*On the saturation effect of kernel ridge regression*, International Conference on Learning Representations, May 2023. Kigali, Rwanda.

Reviewing Service

I regularly and actively serve as a reviewer for conferences including **NeurIPS, ICML, ICLR, AISTATS, AAAI**.

Teaching Experience

2024
- Main Speaker, Seminar for Student Research Training on Kernel Methods, Tsinghua University.

2023
- Excellent Teaching Assistant Scholarship (Tsinghua University) for:
  Multivariate Statistical Analysis (Lecturer: Wanlu Deng),
  Elementary Probability Theory (Lecturer: Wanlu Deng).

Awards and Scholarships

Ph.D. Stage (2022–Present)

2025
- National Scholarship.
- Award for Excellent Poster Presentation at the Ninth Peking-Tsinghua Joint Statistics Symposium.

2024
- Huiyan Scholarship (First Prize Scholarship).

2023
- Excellent Teaching Assistant Scholarship.
- Award for Excellent Poster Presentation at the Seventh Peking-Tsinghua Joint Statistics Symposium.
- Kunhua Scholarship (First Prize Scholarship).

Undergraduate Stage (2018–2022)

2022
- Outstanding Graduate, Tsinghua University.

2021
- Tangjunyuan Scholarship (First Prize Scholarship).

2020
- Hengda Scholarship (First Prize Scholarship).

2019
- First Prize in the Eleventh National Undergraduate Mathematics Contest.
- National Scholarship.

Miscellaneous

- Contributor of TeXify IDEA, an open-soured IntelliJ IDEA plugin for LaTeX support.
- President, Tsinghua Student Reading Club, 2022–2023.